

JKBOSE 2026

CLASS 11TH

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Relations and Functions

MATHS

LECTURE- 07

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

MISCELLANEOUS EXERCISE

CHAPTER – 2

Question

The relation f is defined by $f(x) = \begin{cases} x^2, 0 \leq x \leq 3 \\ 3x, 3 \leq x \leq 10 \end{cases}$

The relation g is defined by $g(x) = \begin{cases} x^2, 0 \leq x \leq 2 \\ 3x, 2 \leq x \leq 10 \end{cases}$

Show that f is a function and g is not a function.

Proof: Given $f(x) = \begin{cases} x^2 ; 0 \leq x \leq 3 \\ 3x ; 3 \leq x \leq 10 \end{cases}$

$$\text{Here } f(3) = 3^2 = 9.$$

$$\& f(3) = 3 \times 3 = 9.$$

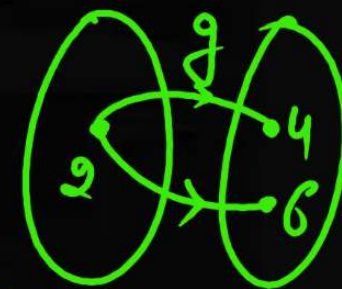
This shows that f is a function.

$$\text{Further } g(x) = \begin{cases} x^2 ; 0 \leq x \leq 2 \\ 3x ; 2 \leq x \leq 10 \end{cases}$$

$$\text{Here } g(2) = (2)^2 = 4$$

$$g(2) = 3 \times 2 = 6$$

$\Rightarrow g$ is not a function.



Question

If $f(x) = x^2$, find $\frac{f(1.1) - f(1)}{(1.1 - 1)}$

(Sol) Given function $f(x) = x^2$

$$\text{So } \frac{f(1.1) - f(1)}{1.1 - 1} = \frac{(1.1)^2 - 1^2}{0.1}$$

$$\Rightarrow \frac{f(1.1) - f(1)}{1.1 - 1} = \frac{1.21 - 1}{0.1} = \frac{0.21}{0.1}$$

$$= \frac{\left(\frac{21}{100}\right)}{\left(\frac{1}{10}\right)} = \frac{21}{10} = 2.1$$

Question

Find the domain of the function $f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}$ (Rational function)

(Sol) Here $x^2 - 8x + 12 = 0$

$$\Rightarrow x^2 - 6x - 2x + 12 = 0$$

$$\Rightarrow x(x - 6) - 2(x - 6) = 0$$

$$\Rightarrow (x - 2)(x - 6) = 0$$

$$\Rightarrow x = 2, 6$$

Therefore the Domain of $f = \mathbb{R} - \{2, 6\}$.

Question

Find the domain and the range of the real function f defined by $f(x) = \sqrt{x-1}$.

(Sol) Given function is;

$$f(x) = \sqrt{x-1}$$

For $f(x)$ to be real, we have

$$x-1 \geq 0$$

$$\Rightarrow x \geq 1$$

$$\text{So, Domain} = [1, \infty)$$

For $x \geq 1$, we have

$$x-1 \geq 0$$

$$\Rightarrow \sqrt{x-1} \geq 0$$

$$\Rightarrow f(x) \geq 0$$

$$\text{So, Range} = [0, \infty)$$

Question

Find the domain and the range of the real function f defined by $f(x) = |x - 1|$.

(Sol) Given function is,

$$f(x) = |x - 1|$$

Since the function is defined for all real values of x , So Domain = $\mathbb{R}$

Further for any real number $x \in \mathbb{R}$

$$|x - 1| \geq 0$$

$$\Rightarrow f(x) \geq 0$$

So Range of $f = [0, \infty)$

Hence the result.

Question

Let $f = \left\{ \left(x, \frac{x^2}{1+x^2} \right) : x \in \mathbb{R} \right\}$ be a function from $\mathbb{R}$ into $\mathbb{R}$. Determine the range of f .

(Sol) Given function is;

$$f = \left\{ \left(x, \frac{x^2}{1+x^2} \right) : x \in \mathbb{R} \right\}$$

$$\Rightarrow f(x) = \frac{x^2}{1+x^2}, x \in \mathbb{R}$$

We know that for any $x \in \mathbb{R}$,
 $x^2 \geq 0$

Dividing $1+x^2$ on b/s and noting that $1+x^2 > 0$

$$\text{we get; } \frac{x^2}{1+x^2} \geq \frac{0}{1+x^2}$$

$$\Rightarrow \frac{x^2}{1+x^2} \geq 0$$

$$\Rightarrow f(x) \geq 0 \quad \text{--- (1)}$$

Further for any $x \in \mathbb{R}$, we have,

$$x^2 < 1+x^2$$

$$\Rightarrow \frac{x^2}{1+x^2} < \frac{\cancel{1+x^2}}{\cancel{1+x^2}}$$

$$\Rightarrow \frac{x^2}{1+x^2} < 1$$

$$\Rightarrow f(x) < 1 \quad \text{--- (2)}$$

From (1) and (2), we get

$$0 \leq f(x) < 1$$

$$\text{So, Range} = [0, 1)$$

Hence the result.

Question

Let $f, g: \mathbf{R} \rightarrow \mathbf{R}$ be defined, respectively by $f(x) = x + 1$, $g(x) = 2x - 3$.

Find $f + g$, $f - g$ and $\frac{f}{g}$ homework

$$(Sol) (f + g)(x) = f(x) + g(x) = x + 1 + 2x - 3$$

$$\Rightarrow (f + g)(x) = 3x - 2.$$

Question

Let $f = \{(1, 1), (2, 3), (0, -1), (-1, -3)\}$ be a function from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = ax + b$, for some integers a, b . Determine a, b .

(Sol) Given function is,

$$f(x) = ax + b$$

Put $x = 0$, we get

$$f(0) = a \cdot 0 + b$$

$$\Rightarrow -1 = 0 + b$$

$$\Rightarrow b = -1$$

$$\text{Also } f(-1) = ax - 1 + b$$

$$\Rightarrow -3 = -a - 1$$

$$\Rightarrow -a = -3 + 1$$

$$\Rightarrow -a = -2$$

$$\Rightarrow a = 2$$

Question

Let R be a relation from $\mathbb{N}$ to $\mathbb{N}$ defined by $R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a = b^2\}$.

Are the following true?

- (i) $(a, a) \in R$ for all $a \in \mathbb{N}$
- (ii) $(a, b) \in R$ implies $(b, a) \in R$
- (iii) $(a, b) \in R, (b, c) \in R$ implies $(a, c) \in R$

Justify your answer in each case.

(Sol) Given Relation is,

$$R = \{(a, b) : a, b \in \mathbb{N} \text{ \& } a = b^2\} \subset \mathbb{N} \times \mathbb{N}.$$

(i) False

Reason: $(2, 2) \notin R$ as $2 \neq 2^2$

(ii) False.

Reason: $(4, 2) \in R$ as $4 = 2^2$
but $(2, 4) \notin R$ as $2 \neq 4^2$.

(iii) False

Reason: $(16, 4) \in R$ as $16 = 4^2$
 $(4, 2) \in R$ as $4 = 2^2$
but $(16, 2) \notin R$ as $16 \neq 2^2$.

Question

Let $A = \{1, 2, 3, 4\}$, $B = \{1, 5, 9, 11, 15, 16\}$ and $f = \{(1, 5), (2, 9), (3, 1), (4, 5), (2, 11)\}$

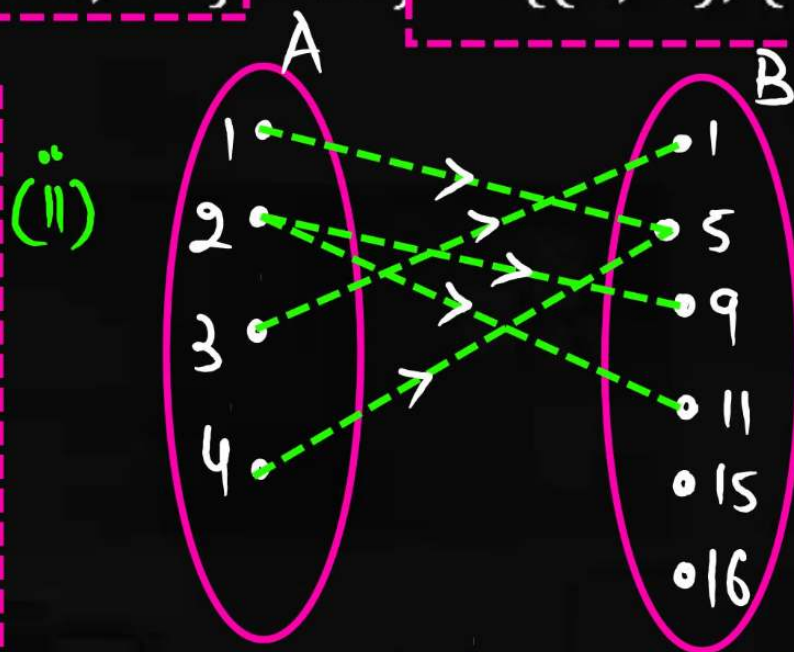
Are the following true?

- (i) f is a relation from A to B
- (ii) f is a function from A to B .

Justify your answer in each case.

(i) Here $f \subset A \times B$

$\Rightarrow f$ is a Relation from A to B .



f is not a function from A to B
as the image of 2 is not unique

Question

Let f be the subset of $\mathbb{Z} \times \mathbb{Z}$ defined by $f = \{(ab, a + b) : a, b \in \mathbb{Z}\}$. Is f a function from $\mathbb{Z}$ To $\mathbb{Z}$?

Justify your answer.

(Sol) Given Relation is,

$$f = \{(ab, a + b) : a, b \in \mathbb{Z}\}$$

$$\Rightarrow f(ab) = a + b \quad \forall a, b \in \mathbb{Z}$$

Put $a = -1, b = -1$, we get;

$$f(-1 \times -1) = -1 - 1$$

$$\Rightarrow f(1) = -2$$

Also, put $a = 1, b = 1$, we get;

$$f(1 \times 1) = 1 + 1 = 2$$

$$\Rightarrow f(1) = 2$$

$\Rightarrow f$ is not a function from $\mathbb{Z}$ to $\mathbb{Z}$.

Question

Let $A = \{9, 10, 11, 12, 13\}$ and let $f : A \rightarrow \mathbb{N}$ be defined by $f(n) =$ the highest prime factor of n . Find the range of f .

(Sol) Given function f is:

$f(n) =$ the highest prime factor of n

$$\text{So, } f(9) = 3$$

$$f(10) = 5$$

$$f(11) = 11$$

$$f(12) = 3$$

$$f(13) = 13$$

$$\text{So Range} = \{3, 5, 11, 13\}$$

$$x - 10$$



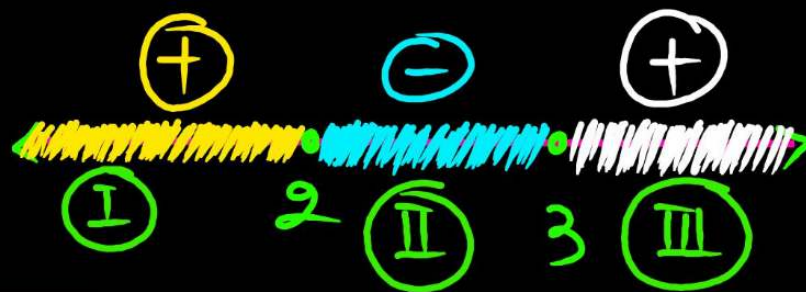
$$x < 10$$

$$\Rightarrow x - 10 < 0$$

$$x > 10$$

$$\Rightarrow x - 10 > 0$$

$$(x - 2)(x - 3)$$



$$\text{Part I: } x < 2 \text{ \& } x < 3$$

$$\Rightarrow x - 2 < 0 \text{ \& } x - 3 < 0$$

$$\Rightarrow (x - 2)(x - 3) > 0$$

$$\text{Part II: } x > 2 \text{ \& } x < 3$$

$$\Rightarrow x - 2 > 0 \text{ \& } x - 3 < 0$$

$$\Rightarrow (x - 2)(x - 3) < 0$$

$$\text{Part III: } x > 3 \text{ \& } x > 2$$

$$\Rightarrow x - 3 > 0 \text{ \& } x - 2 > 0$$

$$\Rightarrow (x - 3)(x - 2) > 0$$

R.D. Sharma (Objective)

Thank You

$$0 \leq f(x) < 1$$

$$[0, 1)$$

* Chapter no: 2
Ends.

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$$\frac{x^2}{1+x^2} = 1$$

$$\cancel{x^2} = 1 + \cancel{x^2}$$

$$0 = 1$$

$$\boxed{f(x) \geq 0}$$

$$\frac{x^2}{1+x^2} < 1$$

$$f(x) = 1$$

$$[0, \infty)$$